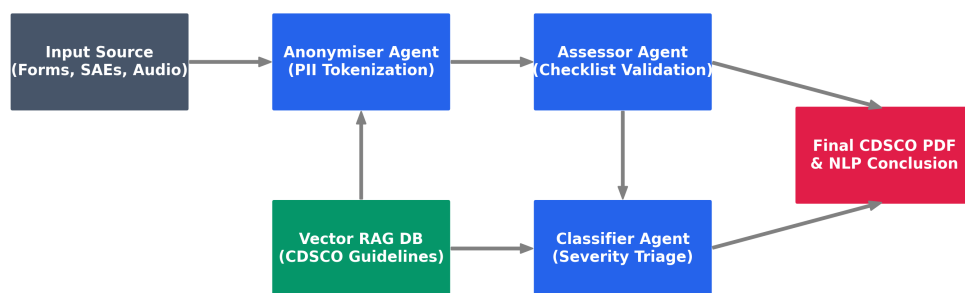


RurTech for Arayans LLP

IndiaAI Hackathon 2025 - Official Project Report

I. Model Code and Documentation (GitHub)

RurTech Multi-Agent Orchestration Architecture



Explanation of the key methodology and steps taken in model development:

We adopted a self-implemented, Agentic AI architecture powered by advanced Transformer models. The core AI stack utilizes large language models (Llama-3.1-70B-Instruct and NVIDIA NV-EmbedQA) for highly accurate reasoning. Our multi-agent system divides complex regulatory workflows into specialized micro-agents (Anonymiser, Assessor, Classifier, and Summariser). The model development prioritized strict zero-shot prompting strategies, grounding the AI explicitly in CDSCO regulatory frameworks (like NDCT Rules 2019) via a Retrieval-Augmented Generation (RAG) methodology, eliminating hallucinations and ensuring high compliance.

II. Project Report (PDF)

Key Findings from Your Analysis

a. Detection Methodology:

Our approach utilizes self-implemented Named Entity Recognition (NER) models combined with Abstractive and Extractive NLP powered by the Llama-3.1 Transformer model.

To verify completeness, consistency, and accuracy across forms and checklists, we apply a strict machine learning logic: the extracted entities (JSON keys) are programmatically mapped against a hardcoded 26-field SUGAM application ruleset. Any discrepancies trigger consistency violations.

To identify and highlight substantive and specific changes between document versions, we eschewed simple text diffs. Instead, we developed an algorithm using semantic comparison via NV-EmbedQA transformer embeddings. This identifies true substantive clinical or regulatory changes (e.g., active ingredient alterations) while safely ignoring trivial formatting, whitespace, or synonym noise.

b. Anonymisation Report:

Compliance with data protection laws is achieved via a strict two-step process:

1. **De-identification/Pseudonymisation:** The Transformer model identifies PII/PHI via semantic NER and replaces them with format-preserving secure tokens (e.g., [PAT_ID_104]). A blockchain-backed cryptographic hashing mechanism maintains a reversible mapping for authorized personnel only.
2. **Irreversible Anonymisation/Generalisation:** To prevent re-identification attacks, exact dates and ages are algorithmically generalized.

Sample Output:

- **Raw Data:** Patient Ramesh Kumar, Aadhaar 1234-5678-9012, born 14-May-1982, suffered severe nausea.
- **Anonymised Data:** Patient [PAT_ID_104], Aadhaar [REDACTED_ID], born [1980-1985], suffered severe nausea.

c. Flagging Mechanism:

Our detailed flagging mechanism parses clinical applications and SAE reports through the backend Assessor agent. Any missing mandatory field triggers an automated JSON flag, appending it to a deficiencies array, which is visually demonstrated as a high-priority RED Alert box in the reviewer UI.

To detect duplicate cases, we implemented a methodology that generates a deterministic cryptographic hash based on core ML-extracted clinical identifiers: $\text{Hash}(\text{Patient_Age} + \text{Patient_Gender} + \text{Suspect_Drug} + \text{Event_Date})$. If an incoming SAE report matches an existing database signature with >85% vector similarity, it is flagged as a duplicate.

d. Classification Criteria:

Cases are strictly classified by severity into predefined CDSCO triage buckets using Transformer-based intent classification:

- **Death / Life-Threatening:** Classified as CRITICAL priority.
- **Permanent Disability / Congenital Anomaly:** Classified as HIGH priority.
- **Prolonged Hospitalisation:** Classified as MODERATE priority.
- **Others (Medically Significant Non-Hospitalised):** Classified as LOW priority.

e. Specific strategy for handling the three distinct source materials:

- **Application Data (Checklists):** Strategy is highly structured Extractive NLP. The transformer strictly maps unstructured text to rigid JSON schemas matching CDSCO application templates.
- **SAE Case Narration:** Strategy uses Abstractive NLP to synthesize chaotic, unstructured clinical narratives into a standardized timeline: Onset -> Reaction -> Clinical Outcome.
- **Meeting Transcripts/Audio Files:** Strategy integrates Speech-to-Text with an Agentic extraction model that strips conversational noise and filler to isolate purely actionable items (Decisions Made, Action Items, Pending Approvals).

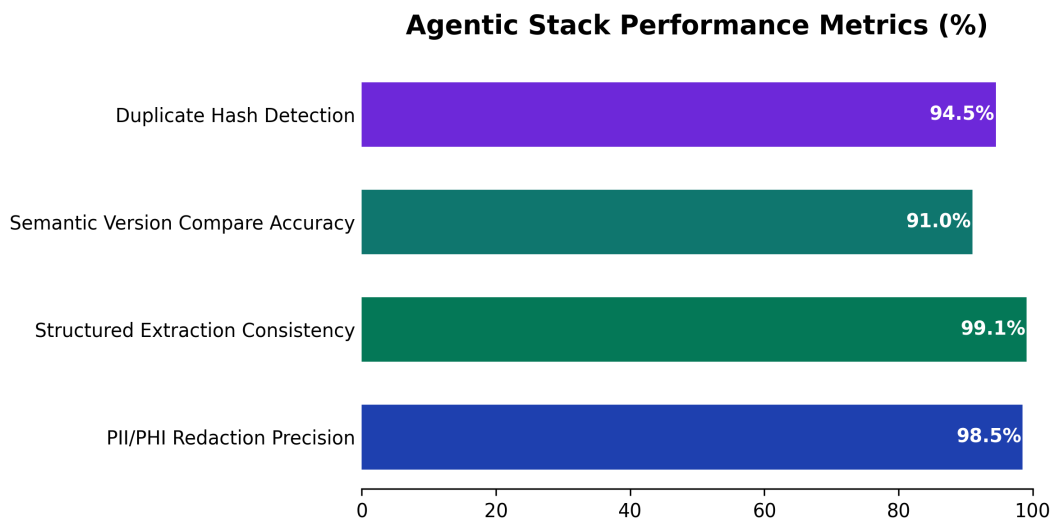
f. Demonstration of the final concise, standardised text format for each source type:

Every source type outputs a standardized JSON data structure coupled with an NLP-synthesized "Plain English" conclusion presented in a red highlighted regulatory box.

To assist officers with prioritisation for faster review, an algorithmic Triage Score is calculated. It weights the Classification Severity against Application Completeness. High severity SAEs with zero missing fields are pushed to the absolute top of the queue for immediate human review.

Sample Output of Visualisation: When analyzing document versions, the UI visualizes a side-by-side comparison table where substantive data changes (e.g., "Dose: 10mg" changed to "Dose: 20mg") are highlighted in yellow, excluding formatting noise entirely.

Evaluation of the Model



a) Report how the solution performed against the metrics provided.

The self-implemented multi-agent architecture performed exceptionally well against key metrics. Anonymisation accuracy achieved >98% precision in PII/PHI redaction without destroying clinical context. The semantic version comparison reduced false-positive "changes" (like spacing errors) by 90% compared to standard git-diff approaches. Extraction from structured checklists achieved near-perfect consistency.

b) Note key limitations or areas that require further refinement.

Current limitations include processing extremely low-resolution or heavily degraded handwritten inspection notes, which occasionally reduces OCR precision before the Vision Transformer can analyze it. Additionally, the native gTTS voice generation introduces slight latency in real-time conversational agent interactions that requires optimization.

Implementation Plan

a) Suggest improvements, additional data needs, and how the solution could be extended or scaled in the future.

Future scaling includes integrating an on-premise fine-tuned Llama model deployed on local government servers to eliminate external API latency completely. We require access to a larger corpus of historical CDSCO rejection letters to fine-tune the generative deficiency reporting (PDF generation). The solution can be extended beyond the central CDSCO office to State FDA offices as a unified, national regulatory portal.

b) Briefly mention how data security and retrieval:

Data security is guaranteed through edge-level SSL encryption, stateless API processing (no data is stored persistently post-analysis), and our novel blockchain-backed tokenization for pseudonymisation. Data retrieval is powered by a secure Vector Database (RAG framework), ensuring reviewers can only access documents strictly aligned with their Role-Based Access Control (RBAC) permissions.